

Education

Duke University - *Sophomore*

Sep 2024 – May 2028

*B.S.E. Mechanical Engineering, Aerospace Engineering & Robotics Certificate; 3.97 GPA***Courses:** Thermodynamics, Fluid Dynamics, Control System, Junior Mechanical Design, Computer Architecture, Advanced Web Development, Constructal Theory, Ethics in Robotics & AI.

Employment

Design Engineer - *Duke DesignHub*

Dec 2025 - Present

- Provide engineering consultation & design services for Duke researchers & hospital clinicians to fulfill their design needs.

Machinist & Teaching Assistant - *Duke Pratt Student Machine Shop*

Sep 2025 - Present

- Support machining classwork, junior & senior design projects by providing machining expertise and shop safety oversight.

Mechanical Engineering Intern - *GEA Group Vietnam - Liquid & Powder Technology*

Apr 2024 - Aug 2024

- Designed & documented basic P&IDs; conducted on-site tank, piping & valves inspections & maintenance for F&B factories.
- Developed general EXCEL calculating tools for GEA's homogenizer flow & heat exchange systems.
- Projects involved: Lothamilk Yogurt Production Line Expansion & Ba Vi Milk Production Line Upgrade

Robotics Research Intern - *Chirikjian Lab, National University of Singapore*

Mar 2023 - Jan 2024

- In charge of Gaussian filter parameter tuning for Computer Vision collision detection, streamlined tuning process by 23%.
- Assisted with post-publication paper reviews on Open Containability and Sitting Affordance via physical simulations.
- Performed 7-DOF robot arm pick & pour sims with MATLAB, ROS-2 & PyBullet, upgraded 1500+ lines codebase to Py3.

University Makerspace Technician - *Maker Innovation Space, VN-UK University*

Dec 2022, Jun 2023, Dec 2023

- Assisted 50+ university student with course projects, ran workshops on 3D Printing & Laser CNC for high school students.
- Designed & constructed $50 \times 50 \text{ cm}^2$ wind tunnel, visualized flow separation & measured different NACA airfoils' performance.

Experience & Projects

Avionics & Separation Lead Engineer, Propulsion Engineer - *Duke AERO*

Jun 2025 - Present

- Direct and oversee Avionics & Separation system development for Duke AERO Rocketry Team's 2025-2026 design cycle.
- Re-engineer tri-blade airbrakes control system for 40% weight reduction and documented detailed system datasheet.
- Develop Monte Carlo framework for characterization of different control algorithms, including gain-scheduling, PID, and adaptive MPC for precise apogee targeting with airbrakes.
- Lead redesign of canards system for 35% control authority increase & 45% reduction in manufacturing & integration time.
- Design first-gen comprehensive bipropellant [liquid engine flight stack architecture](#), support efforts leading up to static hotfire.

Avionics, Structures, Propulsion Design Engineer - *Duke AERO*

Sep 2024 - Jun 2025

- Led [FINSight V2](#): novel dual-layer PCB with embedded accelerometers & DAQ system to characterize rocket fin flutter.
- In charge of design & analysis of IP-67 rated, blast-proof [black box](#) housing & PCB, validated with ANSYS FEA.
- Led research on [canard](#)-tail fin aerodynamic coupling & control surface flutter for canard roll-stabilization system. Developed closed-loop PI ramping control process. Selected for podium presentation at IREC 2025 as an outstanding project.
- Conducted NOX & Kerosene injector simulations on ANSYS Fluent to validate injection pattern and pressure losses.
- Manufactured flight-critical [propulsion hardware](#), including 6" solid rocket motor closures, nozzle & APCP propellant grains.
- Designed modular tank & motor [test stands](#) & DAQ system for cold flow & static fire, withstanding over 35000Ns impulse.
- Assisted with development of LABView-based data logging & control system for ENO.

Technologies

Relevant Skills: ANSYS Structural & Fluent, GD&T, Tolerance Stackup, Precision Machining, Composite Layup & Molding, CAD/CAM (SolidWorks, Onshape, AutoCAD, Fusion360), PCB design (Altium, KiCAD, Eagle), DFM & AM Optimization, Aircraft Design Suite (Athena Vortex Lattice, XFOIL & XFLR5, OpenVSP, MachUp)**Programming, Control Languages & Frameworks:** Python, C, C++, Java, MATLAB, MIPS Assembly, HTML, CSS, JS, TypeScript, Vue, React, Docker, ROS, LABView & NI-DAQmx